

# LAB EXPERIMENT RECORD

## Set 11: Product Category Analysis

|                           |                                         |
|---------------------------|-----------------------------------------|
| Department                | B.Tech AIDS                             |
| Course Code / Course Name | DSA06 / Data Handling and Visualization |
| Experiment                | Data Visualization using R              |
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| Register Number           | 192524110                               |
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| Day                       | Tuesday                                 |

### Aim

To analyze product category sales using a funnel chart, table, and pie chart in R, and to build an interactive R Shiny dashboard combining the pie chart, funnel chart, and table.

### Dataset: Product Categories

| Category    | Sales (in \$) |
|-------------|---------------|
| Electronics | 50000         |
| Clothing    | 35000         |
| Appliances  | 40000         |

**Note:** The dataset gives only total sales for each category, so in the funnel chart each category is shown as one stage, arranged from the highest to the lowest sales, with its percentage of the top stage.

## PROGRAM 1: Funnel Chart of Category Sales

Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
# Create dataset for Set 11
category_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

print(category_data)

# Sort categories from highest to lowest sales (funnel stages)
funnel_data <- category_data[order(-category_data$Sales), ]
funnel_data$Percent <- round(100 * funnel_data$Sales / max(funnel_data$Sales), 1)
print(funnel_data)

# 1. Funnel Chart
n <- nrow(funnel_data)
```

```

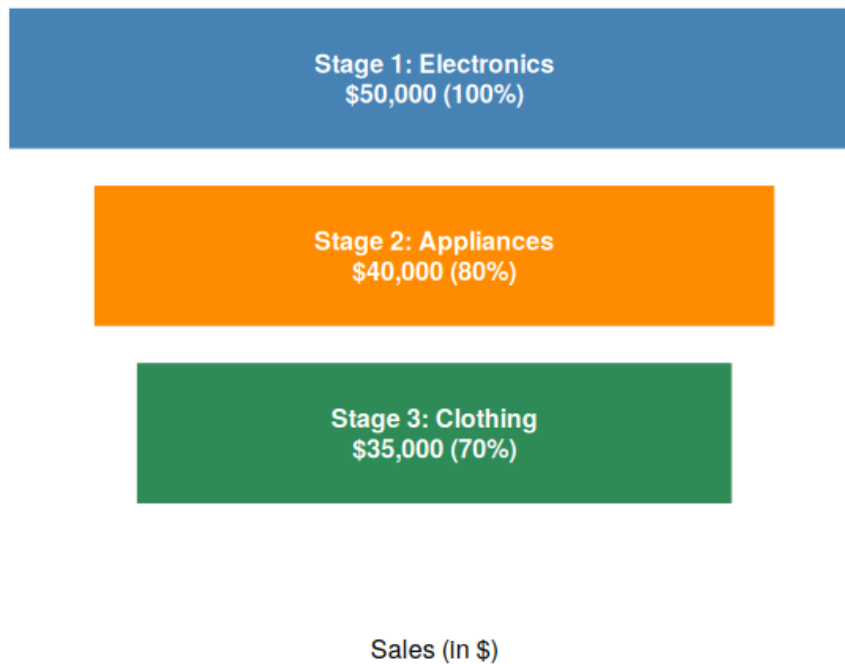
max_sales <- max(funnel_data$Sales)
stage_cols <- c("steelblue", "darkorange", "seagreen")

plot(NA, xlim = c(-max_sales / 2, max_sales / 2) * 1.1, ylim = c(0.5, n + 0.5),
     axes = FALSE, xlab = "Sales (in $)", ylab = "",
     main = "Sales Funnel by Product Category")

for (i in 1:n) {
  y <- n - i + 1
  half <- funnel_data$Sales[i] / 2
  rect(-half, y - 0.4, half, y + 0.4, col = stage_cols[i], border = "white")
  text(0, y, paste0("Stage ", i, ": ", funnel_data$Category[i], "\n$",
                    format(funnel_data$Sales[i], big.mark = ","),
                    " (", funnel_data$Percent[i], "%)"),
       col = "white", font = 2)
}

```

### Sales Funnel by Product Category



Output 1: Sales Funnel by Product Category

**Observation:** Electronics forms the widest stage of the funnel with \$50,000 in sales. Appliances reaches 80% of the Electronics level (\$40,000), and Clothing narrows to 70% (\$35,000). Electronics is the strongest-selling category and Clothing the weakest.

## PROGRAM 2: Table of Category Sales

Build a table to display the sales data for each product category. Label the table elements.

```
# Create dataset for Set 11
category_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

print(category_data)

# 2. Table of sales data for each product category
category_table <- category_data
category_table$Share <- round(100 * category_data$Sales / sum(category_data$Sales), 1)
colnames(category_table) <- c("Category", "Sales ($)", "Share of Total Sales (%)")

cat("=== PRODUCT CATEGORY SALES TABLE ===\n")
print(category_table, row.names = FALSE)
```

```
=== PRODUCT CATEGORY SALES TABLE ===
> print(category_table, row.names = FALSE)
  Category Sales ($) Share of Total Sales (%)
Electronics    50000             40
  Clothing     35000             28
  Appliances    40000             32
>
```

Output 2: Product Category Sales Table

**Observation:** The table lists the sales of each category with its share of total sales (\$125,000). Electronics contributes 40%, Appliances 32%, and Clothing 28%.

## PROGRAM 3: Interactive Dashboard for Product Categories

Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

**Note:** Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 11
category_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
cat_cols <- c(Electronics = "steelblue", Appliances = "darkorange", Clothing = "seagreen")

ui <- fluidPage(
  titlePanel("Product Category Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("cat", "Category:", choices = names(cat_cols), selected =
names(cat_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("piePlot", height = "300px")),
        column(6, plotOutput("funnelPlot", height = "300px"))
      ),
      h4("Product Category Sales Table"),
      tableOutput("catTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(category_data, Category %in% input$cat) })

  output$piePlot <- renderPlot({
    d <- filtered()
    pct <- round(100 * d$Sales / sum(d$Sales), 1)
    pie(d$Sales, labels = paste0(d$Category, " (", pct, "%)", col = cat_cols[d$Category],
      main = "Sales Distribution by Category")
  ), width = 420, height = 300, res = 96)

  output$funnelPlot <- renderPlot({
    d <- filtered()
    d <- d[order(-d$Sales), ]
    n <- nrow(d)
    plot(NA, xlim = c(-30000, 30000), ylim = c(0.5, 3.5), axes = FALSE,
      xlab = "Sales (in $)", ylab = "", main = "Sales Funnel by Category")
    for (i in seq_len(n)) {
      y <- 3 - i + 1
      rect(-d$Sales[i] / 2, y - 0.4, d$Sales[i] / 2, y + 0.4,
        col = cat_cols[d$Category[i]], border = "white")
      text(0, y, paste0(d$Category[i], "\n$", format(d$Sales[i], big.mark = ",")),
```

```

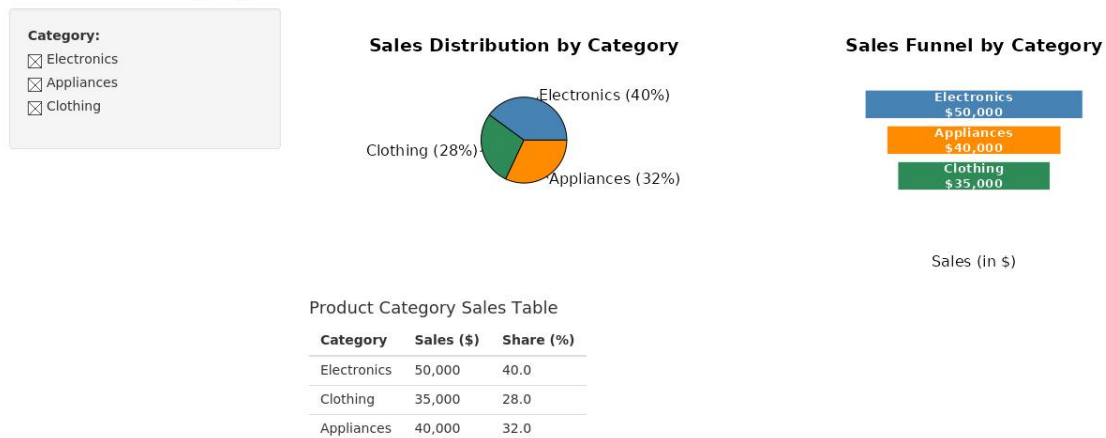
      col = "white", font = 2, cex = 0.85)
    }
  }, width = 420, height = 300, res = 96)

output$catTable <- renderTable({
  d <- filtered()
  d$Share <- format(round(100 * d$Sales / sum(d$Sales), 1), nsmall = 1)
  d$Sales <- formatC(d$Sales, format = "d", big.mark = ",",)
  colnames(d) <- c("Category", "Sales ($)", "Share (%)")
  d
})
}

shinyApp(ui = ui, server = server)

```

### Product Category Dashboard



**Output 3: Product Category Interactive Dashboard (R Shiny)**

**Observation:** The dashboard shows the sales pie chart, the sales funnel, and the category sales table together. Unticking a category in the Category filter removes it from all three views, and the percentage shares are recalculated for the selected categories.

## PROGRAM 4: Pie Chart of Sales by Category

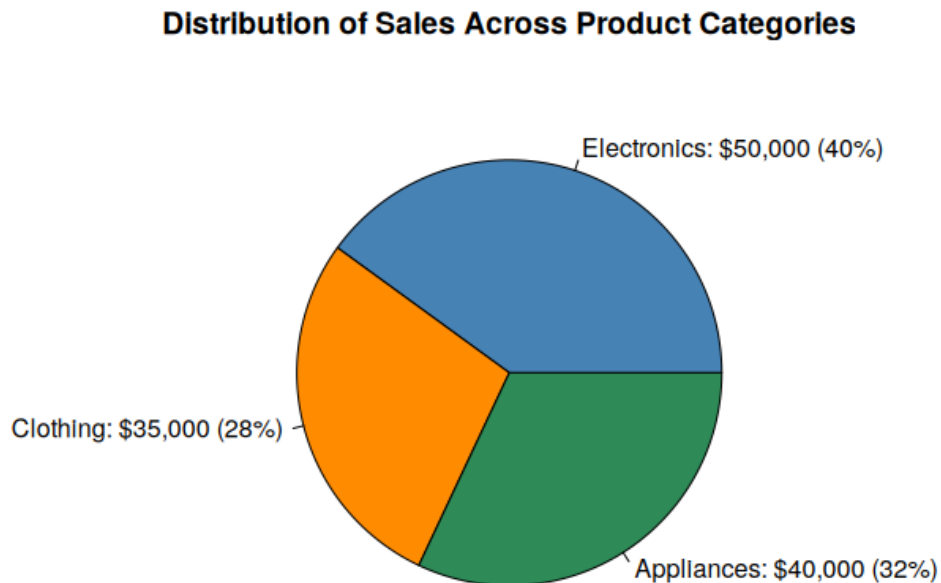
Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
# Create dataset for Set 11
category_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

print(category_data)

# Percentage share of each category
sales_pct <- round(100 * category_data$Sales / sum(category_data$Sales), 1)
pie_labels <- paste0(category_data$Category, ": $",
  format(category_data$Sales, big.mark = ","),
  " (", sales_pct, "%)")

# 4. Pie Chart of Sales by Product Category
pie(category_data$Sales,
  labels = pie_labels,
  col = c("steelblue", "darkorange", "seagreen"),
  main = "Distribution of Sales Across Product Categories")
```



Output 4: Distribution of Sales Across Product Categories

**Observation:** Electronics takes the largest slice of the pie (40% of total sales), followed by Appliances (32%) and Clothing (28%). The shares are fairly balanced, with no single category making up more than half of the sales.



## Procedure

1. Create the category data frame with Category and Sales.
2. Sort the categories by sales using `order()` and draw the funnel stages using `rect()` and `text()`.
3. Calculate the share of total sales, rename the columns with `colnames()`, and print the labelled table.
4. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
5. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.
6. Calculate percentage shares and use `pie()` to draw the labelled pie chart.

## Result

The product category data was successfully visualized using a funnel chart, table, and pie chart in R, and an interactive R Shiny dashboard was built combining the pie chart, funnel chart, and table.

### Key R Functions Used

| Function                                                     | Purpose                                        |
|--------------------------------------------------------------|------------------------------------------------|
| <code>data.frame()</code>                                    | Creates the product category dataset.          |
| <code>order()</code>                                         | Sorts categories from highest to lowest sales. |
| <code>rect()</code>                                          | Draws the funnel stages.                       |
| <code>text()</code>                                          | Labels each funnel stage.                      |
| <code>format()</code>                                        | Formats sales values with commas.              |
| <code>pie()</code>                                           | Creates the sales distribution pie chart.      |
| <code>fluidPage()</code> / <code>checkboxGroupInput()</code> | Creates the dashboard layout and filters.      |
| <code>reactive()</code> / <code>renderPlot()</code>          | Filters the data and redraws the charts.       |
| <code>renderTable()</code>                                   | Displays the filtered table on the dashboard.  |
| <code>shinyApp()</code>                                      | Runs the interactive dashboard.                |